2022-2 Calculus II (06)

final Exam

2022.12.20, 15:00–16:30

10 problems. All your solutions should contain rigorous explanation.

1. Evaluate the triple integral
$$\iiint_T y\,dV,$$
where T is the tetrahedron bounded by the planes $x+2y+z=2$, $x=2y$, $x=0$, and $z=0$.

2. Evaluate the double integral
$$\iint_R (x-y)^2 e^{y^2-x^2}\,dA,$$
where R is the trapezoidal region with vertices $(1,0)$, $(2,0)$, $(0,-2)$, and $(0,-1)$.

3. (10pts) Let $\mathbf{F}(x,y) = \dfrac{y\,\mathbf{i} - x\,\mathbf{j}}{x^2+y^2}$. Evaluate the line integrals
$$\int_C \mathbf{F}\cdot d\mathbf{r} \quad \text{and} \quad \int_C \mathbf{F}\cdot\mathbf{n}\,ds,$$
where C is the counterclockwise-oriented circle with radius 2 and center the origin, and $\mathbf{n}$ is the outward unit normal vector to C.

4. (10pts) Let $\mathbf{F}(x,y,z) = \sin y\,\mathbf{i} + (x\cos y + \cos z)\,\mathbf{j} + (1 - y\sin z)\,\mathbf{k}$ and C is the curve $x = \sin t$, $y = t$, $z = 2t$, $0 \le t \le \pi$. Evaluate the line integral
$$\int_C \mathbf{F}\cdot d\mathbf{r}.$$

5. (10pts) Evaluate
$$\oint_C y^2\,dx + 3xy\,dy,$$
where C is the boundary of the region D in the upper-half plane between the ellipses by $x^2+4y^2=4$ and $x^2+4y^2=16$

6. (10pts) Let $\mathbf{F}(x,y) = P(x,y)\,\mathbf{i} + Q(x,y)\,\mathbf{j}$. Suppose that D is a bounded and simply connected region in $\mathbb{R}^2$, the boundary of D, denoted by C, is smooth and given by the vector equation $\mathbf{r}(t) = x(t)\,\mathbf{i} + y(t)\,\mathbf{j}$, $a \le t \le b$, and P and Q have continuous partial derivatives in an open region containing D. **Using Green's theorem**, prove that

$$\oint_C \mathbf{F} \cdot \mathbf{n}\, ds = \iint_D \operatorname{div} \mathbf{F}(x,y)\, dA,$$

where $\mathbf{n}$ is the outward unit normal vector given by $\frac{y'(t)}{|\mathbf{r}'(t)|}\,\mathbf{i} - \frac{x'(t)}{|\mathbf{r}'(t)|}\,\mathbf{j}$.

7. (10pts) Find the area of the surface $z = \sqrt{2xy}$, $1 \le x \le 4$, $1 \le y \le 4$.

8. (10pts) Let $\mathbf{F}(x,y,z) = (x + x^2)\,\mathbf{i} + (y - 2xy)\,\mathbf{j} - 3z\,\mathbf{k}$. Suppose S is the boundary of the region $R = \{(x,y,z) : -1 \le x \le 1,\ -1 \le y \le 1,\ -1 \le z \le 1\}$ **without the lid** $S_0 = \{(x,y,1) : -1 \le x \le 1,\ -1 \le y \le 1\}$, oriented outward to R. Evaluate the surface integral

$$\iint_S \mathbf{F} \cdot d\mathbf{S}.$$

9. (10pts) Let $\mathbf{F}(x,y,z) = 2x^2\,\mathbf{i} + y^2\,\mathbf{j} + z^2\,\mathbf{k}$, and S be the boundary of the solid half-cylinder $0 \le z \le \sqrt{1 - y^2}$, $0 \le x \le 2$. Evaluate the surface integral

$$\iint_S \mathbf{F} \cdot d\mathbf{S}.$$

10. (10pts) Evaluate the line integral

$$\int_C (y + \sin x)\, dx + (z^2 + \cos y)\, dy + \frac{4}{3}x^3\, dz$$

where C is the curve $\mathbf{r}(t) = \langle \sin t, \cos t, \sin 2t \rangle$, $0 \le t \le 2\pi$.